

# ex2 A114290047

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## R Markdown

```
# Commute data
commute <- c(18, 25, 32, 41, 27, 35, 22, 19, 45, 38, 29, 24,
            31, 36, 40, 26, 33, 28, 21, 44, 30, 37, 23, 39)

# Find minimum and maximum
min(commute)

## [1] 18

max(commute)

## [1] 45

# Find the range
range <- max(commute) - min(commute)
range

## [1] 27

# Determine class width
class_width <- ceiling(range / 5)
class_width

## [1] 6

# Create 5 classes
classes <- cut(commute, breaks = 5)

# Frequency of each class
frequency <- table(classes)
frequency

## classes
##   (18,23.4] (23.4,28.8] (28.8,34.2] (34.2,39.6] (39.6,45]
##           5           5           5           5           4

# Cumulative frequency
cumulative_frequency <- cumsum(frequency)
cumulative_frequency

##   (18,23.4] (23.4,28.8] (28.8,34.2] (34.2,39.6] (39.6,45]
##           5           10          15          20          24

# Create grouped frequency distribution
frequency_distribution <- data.frame(
```

```

Class = names(frequency),
Frequency = as.vector(frequency),
Cumulative_Frequency = as.vector(cumulative_frequency)
)

# Display the final table
frequency_distribution

##           Class Frequency Cumulative_Frequency
## 1  (18,23.4]           5                    5
## 2 (23.4,28.8]           5                   10
## 3 (28.8,34.2]           5                   15
## 4 (34.2,39.6]           5                   20
## 5  (39.6,45]           4                   24

# Data
commute <- c(18, 25, 32, 41, 27, 35, 22, 19, 45, 38, 29, 24,
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```

```

# Install ggplot2 if needed
install.packages("ggplot2")

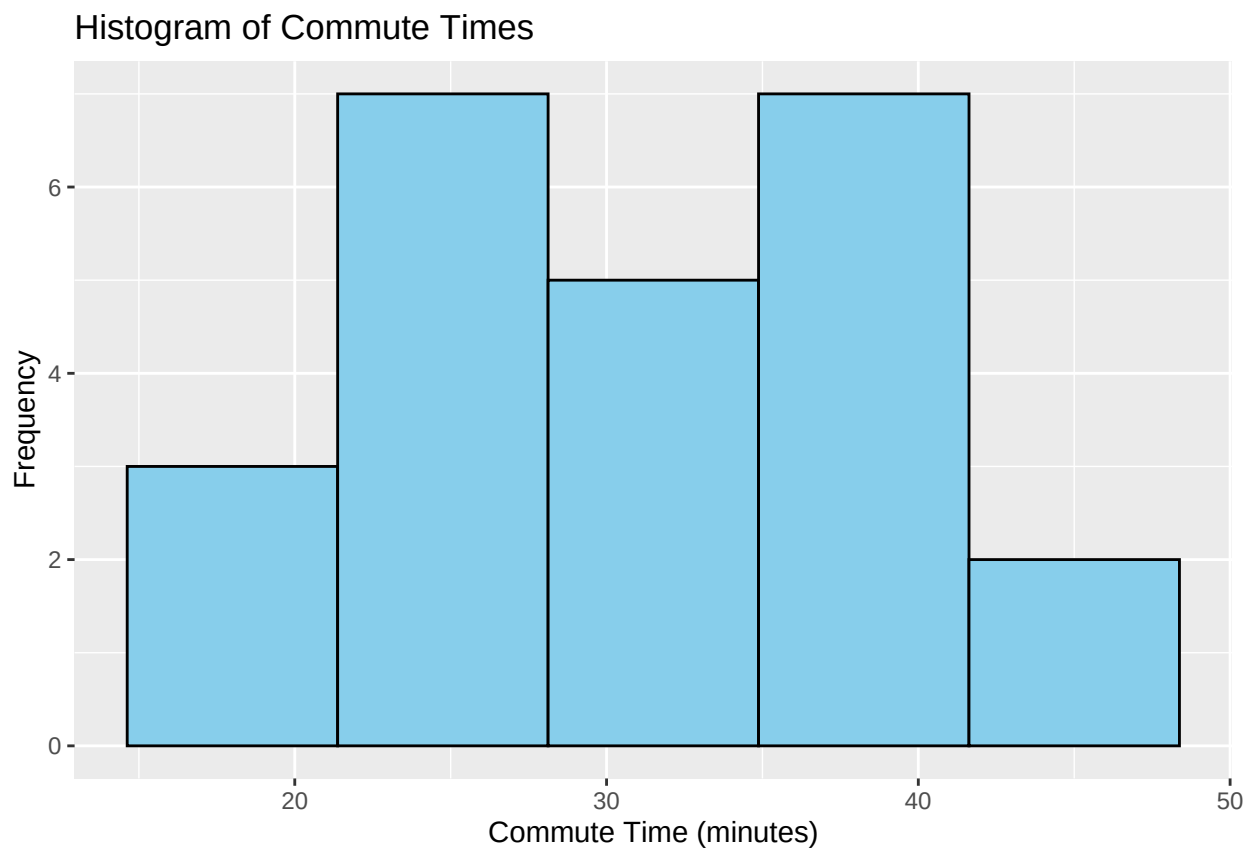
## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.6'
## (as 'lib' is unspecified)

# Load ggplot2
library(ggplot2)

# Commute data
commute <- c(18, 25, 32, 41, 27, 35, 22, 19, 45, 38, 29, 24,
            31, 36, 40, 26, 33, 28, 21, 44, 30, 37, 23, 39)

# Draw histogram
ggplot(data.frame(commute), aes(x = commute)) +
  geom_histogram(bins = 5, color = "black", fill = "skyblue") +
  labs(
    title = "Histogram of Commute Times",
    x = "Commute Time (minutes)",
    y = "Frequency"
  )

```



```

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# Commute data
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```

```

# Minimum and maximum
min(commute)

## [1] 18

max(commute)

## [1] 45

# Range
range <- max(commute) - min(commute)
range

## [1] 27

# Class width
class_width <- ceiling(range / 5)
class_width

## [1] 6

# Create class boundaries
breaks <- seq(18, 48, by = 6)

# Create classes
classes <- cut(commute, breaks = breaks, right = FALSE)

# Frequency
frequency <- table(classes)
frequency

## classes
## [18,24) [24,30) [30,36) [36,42) [42,48)
##      5      6      5      6      2

# Cumulative frequency
cumulative_frequency <- cumsum(frequency)
cumulative_frequency

## [18,24) [24,30) [30,36) [36,42) [42,48)
##      5      11      16      22      24

# Upper class boundaries
upper_boundaries <- breaks[-1]

# Create ogive data
ogive_data <- data.frame(
  Upper_Boundary = upper_boundaries,
  Cumulative_Frequency = as.numeric(cumulative_frequency)
)

# Display ogive data
ogive_data

##   Upper_Boundary Cumulative_Frequency
## 1             24                  5
## 2             30                 11
## 3             36                 16
## 4             42                 22

```

## 5

48

24

```
# Draw ogive  
ggplot(ogive_data, aes(x = Upper_Boundary, y = Cumulative_Frequency)) +  
  geom_line() +  
  geom_point(size = 3) +  
  labs(  
    title = "Ogive of Commute Times",  
    x = "Upper Class Boundary (minutes)",  
    y = "Cumulative Frequency"  
  )
```

